

7. Project Documentation Phase

Project: PocketSmart AI - Smart Budget & Recommendation Assistant

7.1 Introduction to Project Documentation

Purpose of Documentation

The Project Documentation Phase organizes the information needed to understand, maintain, explain, and evaluate PocketSmart AI. The project documentation describes its purpose, architecture, workflow, technologies, activities, user pages, and conclusion.

Project Overview

PocketSmart AI is described as a GenAI-powered, cross-platform recommendation system that provides personalized, budget-based suggestions for products and services in areas such as home interiors, party planning, and jewelry.

Problem and Solution

The documentation records the main problem as the difficulty of managing budgets across different products, platforms, and price ranges. The proposed solution combines user preferences and budgets with a Gemini-based recommendation engine.

Main Use Cases

The documented use cases are Home Interior Planning, AI-Based Party Budget Planning, and Jewelry Recommendations for Occasions. Each use case has its own inputs and recommendation behavior.

Documentation Structure

The source project is organized around architecture, prerequisites, workflow activities, milestones, frontend pages, testing, and a conclusion. This structure provides a clear reference for explaining the complete application.

Page 2 - Technical Documentation

Architecture Documentation

The architecture section describes a modular platform that connects the frontend, FastAPI backend, Gemini AI layer, and external product or service sources.

Technology Documentation

The project identifies FastAPI for backend services, Gemini 1.5 Flash Pro for AI processing, third-party APIs or simulated sources for product and service data, and HTML/CSS with JavaScript for the frontend.

Backend Documentation

Backend documentation includes routes for Home, Party, and Jewelry generation, authentication, session information, recommendation details, history, and startup behavior.

Modular Code Documentation

The documented structure separates routes, services, and models. The recommendation utility area handles prompt orchestration, budget formatting, domain segmentation, and image analysis.

Configuration Documentation

The workflow records API access, key configuration, usage limits, environment settings, and validation of Gemini connectivity. These details are important for reproducing the project setup.

Page 3 - User Interface Documentation

Home Page

The Home Page is documented as the main landing page introducing PocketSmart AI's features and providing access to planner modules and user actions.

Authentication Pages

Register and Login pages are documented for account creation and authentication. Logout terminates the active user session.

Dashboard

The User Dashboard is documented as a place to display recent recommendations, saved queries, and personalized insights.

Planner and Recommendation Pages

Separate planner and recommendation pages are documented for Home Interior, Party, and Jewelry use cases. They support input collection and structured display of AI-generated suggestions.

History and Supporting Pages

The Recommendation History page records previous queries and results. The project also documents a Testimonials Page and footer content as part of the frontend.

User Flow Documentation

A user selects a planner, enters the required information, submits the request, receives AI-generated recommendations, and can review results through the relevant recommendation or history pages.

Page 4 - Process and Testing Documentation

Project Workflow

The documented workflow moves from Gemini AI setup to core functionality development, backend implementation, UI development, and testing and optimization.

Activity Documentation

Activities include API setup, prompt configuration, backend routing, planner development, authentication, CORS and session handling, responsive forms, dynamic result display, and testing.

Testing Documentation

The testing section records real-world scenarios across Home, Party, and Jewelry categories and identifies response quality, budget adherence, platform accuracy, prompt optimization, validation, and fallback behavior as evaluation areas.

Maintenance Documentation

The modular structure provides a useful maintenance reference because routes, services, models, and planner-specific functions are kept organized.

Limitations and Notes

The source document mentions mock API calls or simulated sourcing for external platforms during development. Therefore, such data-source behavior should be described clearly when the project is presented or reused.

Page 5 - Final Documentation Package

Recommended Documentation Set

A complete project documentation package should contain the project overview, problem statement, planner descriptions, architecture, technology stack, backend routes, authentication flow, frontend pages, testing activities, and final conclusion.

Implementation Reference

The documented workflow can be used as a reference when explaining how Gemini API access, FastAPI routes, planner modules, UI pages, and testing were implemented.

Demonstration Support

The documentation should also provide enough information to explain a live demonstration, including the expected inputs for each planner and the type of recommendation output produced.

Future Maintenance Reference

Keeping the architecture and route information documented helps future developers identify where to update planner logic, prompts, validation, external data sourcing, or interface components.

Conclusion

The Project Documentation Phase brings the technical and functional information of PocketSmart AI into a structured reference. It supports development, testing, maintenance, evaluation, and demonstration by clearly describing what the system does and how its major components work together.